

manufacturing the products. This would reduce the cost of development and risk of failure.

### Learning: A long way to go

The pandemic made every student realise that learning through online platforms was not the same as being in school with friends, and they are delighted to be back on campus. However, educationalists still worry about what would happen if another such situation arises where kids are unable to go to school. As a workaround, experts suggest that schools attempt a multi-modal education even on a normal day, so that they can quickly transition to a digital mode when required.

While schools struggled for continuity using e-learning platforms, videos, and chatrooms, in the future the metaverse could be a more effective option. An immersive learning experience with graphics-intensive virtual platforms will empower students to understand concepts better, and to experience how certain things work in the real world. It might be difficult for children to understand how a bank works. What if they could enter a simulated environment and experience it for themselves? The metaverse makes that possible—and it might be a great learning aid even on a normal day.

However, the fact remains that the metaverse is still in a very nascent stage and its constituent technologies are too expensive to have a wide reach. Where it could work wonders is in higher education, upskilling, and corporate training.

Universities such as the University of Nevada at Las Vegas and the University of Miami are experimenting with immersive learning experiences. The University of Miami's XR Initiative spans subjects like architecture, healthcare, climate change and behavioural research. Medical students, for example, can learn how to administer anaesthesia to a virtual patient in a simulated operating room.



Figure.2: BMW Group uses augmented reality in prototyping (Courtesy: BMW Group)

Invact Metaversity, founded by former Twitter India head Manish Maheshwari and former Microsoft engineer Tanay Pratap, is a full virtual reality university campus replete with a library, cafeteria, hallways, and a playground. It currently offers a 16-week management course, which students can attend from anywhere using a VR headset. It can bring together students and teachers from around the world. The company claims that students can also enjoy the social nuances of college life in the virtual world.

### Upskilling: See you at the top!

According to a PwC report, VR is already known to be effective for teaching hard skills and for job skills simulations, such as a flight simulator to train pilots. However, a survey done by them during the pandemic times showed that VR was, surprisingly, very helpful in soft skills training too! The survey showed that VR can help business leaders upskill their employees faster, even at a time when training budgets may be shrinking and in-person training may be off the table, as people continue to observe social distancing. Learners trained with VR

were up to 275% more confident to act on what they learned after training—a 40% improvement over in-person classroom learning and a 35% improvement over eLearning. It showed that employees could also be trained up to four times faster than in a classroom. No wonder then that companies are also turning to the metaverse to train and upskill employees.

Recently, Accenture announced that its 150,000 new hires this fiscal year will use VR headsets to work from the metaverse on their first day. They will meet, collaborate, and learn on the Nth Floor, the company's virtual campus on the metaverse. Accenture believes this immersive experience is 30-40% more productive than traditional ways of learning.

Earlier this year, Mumbai based XR hardware and software company AjnaLens joined hands with Tata Technologies to digitise 150 industrial training institutes (ITIs) in Karnataka. The ITIs will be equipped with AjnaXR stations to provide immersive skill training. The made-in-India, highly-scalable VR training simulator virtually teleports a trainee to an actual jobsite and provides multi-lingual training. AjnaLens has